Data Visualization & Analytics

Factors Affecting Crime

Optimizing Law Enforcement Resource Allocation

Sector	Public Safety & Law Enforcement
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GitHub	github.com/codeewizard/SectionA_G15_FactorsAffectingCrime
Tableau	https://public.tableau.com/app/profile/apoorva.choudhary3398/viz/Factors_Affecting_Crime/CrimeOverviewandTrends?publish=yes

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Apoorva	Project Lead
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1. Abstract

This report presents a complete data-driven investigation of crime patterns in Los Angeles (2020-2024) using the LAPD Crime Incidents dataset. The analysis combines a Python-based ETL pipeline, exploratory data analysis, statistical hypothesis testing, machine learning models, and an interactive three-dashboard Tableau suite. The central objective is to identify when, where, and under what victim-context conditions crime concentration is highest, and convert those findings into actionable resource-allocation recommendations for law enforcement decision-makers.

2. Executive Summary

Problem. Urban law enforcement agencies face finite budgets and severe personnel constraints. Deploying officers uniformly across unequal risk zones wastes resources in safe areas while high-crime hotspots remain understaffed. The LAPD, overseeing one of the most complex urban environments in the United States, requires a data-backed deployment framework.

Approach. Raw LAPD data (1,004,894 records) was cleaned to 326,327 high-fidelity rows using a Python ETL pipeline. EDA, chi-square tests, ANOVA, z-score anomaly detection, logistic regression, and random forest classification were applied. Results were visualised across three interactive Tableau dashboards filterable by crime type, hour, month, and victim age group.

Key Findings:

- Crime volume peaks at 8 PM (20:00), concentrating 5.75% of all incidents in a single hour.
- Two divisions - 77th Street (24,124) and Central (20,291) - account for nearly a fifth of citywide crime.
- 45.5% of all incidents are violent (assault/battery), driving the need for physical-response units.
- Only 17.28% of cases are resolved; 58.34% remain under active investigation.

Key Recommendations:

- Reallocate 10% of patrol units from low-risk divisions to 77th Street and Central during 19:00-21:00.
- Restructure graveyard-shift staffing to reflect the 30.91% night crime ratio.
- Introduce case-triage protocols to lift the critically low 17.28% resolution rate.

3. Sector & Business Context

Public safety is one of the most resource-constrained sectors in municipal government. Every dollar misallocated to low-risk patrol zones is a dollar unavailable at a genuine hotspot. The LAPD manages 21 geographic divisions across a sprawling metropolitan area where crime is highly non-uniform in time, space, and type.

The rise of data-driven policing, accelerated by the availability of open incident datasets and modern BI tooling, has created an opportunity to move from reactive dispatch to proactive deployment. Decision-makers including shift lieutenants, resource planners, and city councils can now base patrol schedules on empirical risk signals rather than administrative tradition. This project directly serves that transition, converting five years of LAPD incident records into a deployable intelligence layer.

Business value. By concentrating resources at the right place and time without increasing headcount or budget, the department can achieve measurably higher coverage efficiency, faster response times, and improved case clearance - all leading indicators of public safety outcomes.

4. Problem Statement & Objectives

Formal Problem Definition. What temporal, spatial, and victim-context factors most significantly influence crime occurrence patterns across LAPD divisions from 2020-2024, and how can these patterns be operationalised to optimise law enforcement resource allocation?

Objectives:

- Build a reliable end-to-end Python analytics pipeline from raw LAPD data to reporting-ready outputs.
- Identify temporal, spatial, and victim-context factors linked with crime volume and concentration.
- Validate patterns through statistical methods (chi-square, ANOVA, t-tests, z-score).
- Build baseline predictive models for binary violent-crime classification and crime-type prediction.
- Prepare aggregated KPI indicators and an interactive Tableau dashboard suite for operational use.

In Scope: LAPD crime incidents 2020-2024; temporal, spatial, and victim dimensions; Tableau dashboards; Python-based statistical and ML analysis. **Out of Scope:** Real-time data feeds, external socioeconomic datasets, weather integration (identified as future work).

Success Criteria: Statistically validated patterns confirmed by p-value testing; actionable recommendations with quantified expected impact; Tableau dashboard accessible publicly with full interactive filtering.

5. Data Description

Dataset Profile

Attribute	Detail
Source	LAPD Crime Incidents Dataset (data.lacity.org)
Raw Records	1,004,894 rows x 28 columns
Cleaned Records	326,327 rows x 24 columns
Time Period	January 1, 2020 - December 30, 2024
Spatial Coverage	21 LAPD area divisions
Unit of Analysis	One row per incident report (DR_NO)
Format	CSV

Key Columns

Column	Description	Role in Analysis
DATE OCC	Date crime occurred	Monthly/yearly trends, KPI computation
TIME OCC	Time of occurrence	Peak-hour index, time-of-day bucketing
AREA NAME	LAPD division name	Spatial concentration, hotspot identification
Crm Cd Desc	Crime type description	Category analysis, violent crime flag
Vict Age	Victim age in years	Age group segmentation, demographic KPIs
Vict Sex	Victim sex code (F/M/X)	Gender split filter

Column	Description	Role in Analysis
Vict Descent	Victim ethnicity code	Demographic breakdown
LAT / LON	Incident coordinates	Hotspot map in Tableau
Status Desc	Case disposition	Resolution rate KPI
Part 1-2	Crime severity grouping	Severity split analysis

Data Limitations

The dataset captures only reported incidents; systemic under-reporting in marginalised communities means true crime volume is higher than measured. December 2024 data is partial and biases naive year-end comparisons. External variables known to influence crime - live weather data, socioeconomic indicators, and city event schedules - are absent from this analysis.

6. Data Cleaning & ETL Pipeline

The Python ETL pipeline processes raw data through five sequential stages, executed in Jupyter notebooks committed to GitHub. The pipeline reduces 1,004,894 raw rows to 326,327 high-fidelity records ready for analysis.

Stage 1 - Extraction (01_extraction.ipynb)

Loads the raw LAPD CSV and performs initial schema inspection, shape reporting, and documentation of data quality issues for downstream treatment.

Stage 2 - Cleaning (02_cleaning.ipynb)

Step	Action	Result
Date/Time Standardization	Converted DATE OCC and Date Rptd to datetime; formatted TIME OCC	Consistent temporal types across all records
Deduplication	Dropped exact duplicates on DR_NO	No duplicate incident records in final dataset
Missing Value Removal	Applied dropna() after targeted column pruning	Row count reduced from 1,004,894 to 326,327
Coordinate Filtering	Removed rows with missing LAT/LON values	All remaining records spatially mappable
Victim Age Imputation	Replaced invalid ages (<=0) with median = 35; 26,084 records corrected	Valid age range maintained throughout dataset
Column Pruning	Dropped sparse columns: Crm Cd 2/3/4, Cross Street (>80% missing)	Column count reduced from 28 to 24
Category Normalization	Mapped unknown Vict Sex and Descent codes to "X"	Clean, consistent categorical fields for analysis

Stage 3 - EDA (03_eda.ipynb)

Univariate and bivariate analysis across temporal, spatial, and victim dimensions. IQR outlier detection identified 4,348 victim age outliers above 74.5 years and confirmed no significant outliers in the hourly crime distribution.

Stage 4 - Statistical Analysis (04_statistical_analysis.ipynb)

Hypothesis testing, ML modelling, anomaly detection, and seasonal decomposition (detailed in Section 9).

Stage 5 - Final KPI Prep (05_final_load_prep.ipynb)

Produces aggregated tables for all dashboard KPIs: volume, temporal, spatial, category, and enforcement-context metrics.

7. KPI & Metric Framework

326,327	30.91%	179	31.06%
Total Crimes	**Night Crime Ratio**	**Avg Crimes/Day**	**Weekend Ratio**
2020-2024 baseline	21:00-05:00	Daily volume	Sat + Sun

45.5%	35.8%	17.28%	58.34%
Violent Crime Ratio	**Top 5 Area Conc.**	**Resolution Rate**	**Pending Rate**
Assault/battery	5 divisions	Closed cases	Under investigation

KPI	Formula	Value	Business Relevance
Total Crimes	COUNT(DR_NO)	326,327	Baseline for all resource allocations
Night Crime Ratio	night_crimes / total x 100	30.91%	Drives graveyard-shift scheduling logic
Peak Hour Index	peak_hour_crimes / total x 100	5.75%	Justifies dedicated 8 PM surge deployment
Violent Crime Ratio	is_violent / total x 100	45.5%	Dictates response unit type required
Top 5 Area Conc.	top5_sum / total x 100	35.8%	Justifies geographic prioritization
Avg Victim Age	MEAN(Vict Age)	37.26 yrs	Informs community program targeting
Resolution Rate	resolved / total x 100	17.28%	Measures investigative effectiveness
Pending Rate	Invest_Cont / total x 100	58.34%	Signals severity of case backlog
Weekend Crime Ratio	weekend_crimes / total x 100		Weekend staffing benchmark
Crime Diversity Index	COUNT(DISTINCT crime_type)	122	Top categories driving total volume

8. Exploratory Data Analysis

Temporal Analysis

Monthly crime counts from 2020-2024 reveal a stable band of 4,500-6,000 incidents per month, with a measurable Q3 seasonal peak recurring annually. The single-hour peak at 20:00 (8 PM) accounts for 18,765 incidents, which is 5.75% of all crime, a concentration with no parallel in any other hour. The mean crime hour is 13.27 (early afternoon) while the modal hour is 20:00, indicating a strongly skewed evening concentration. The weekend crime ratio of 31.06% exceeds the expected 28.6% baseline (2/7 days), confirming disproportionate weekend risk.

Spatial Analysis

Crime is heavily geographically concentrated. The top five LAPD divisions - 77th Street (24,124), Central (20,291), Southeast (19,378), Southwest (17,275), and Hollywood (16,033) - together represent 35.8% of citywide volume. The bottom divisions (Foothill, West LA, Devonshire) consistently record the lowest volumes, confirming the viability of resource reallocation from low-risk to high-risk zones. The crime hotspot map confirms spatial clustering in central and south-central LA, with clear intensity gradients visible in the Tableau heatmap.

Crime Type Distribution

Assault-linked offences dominate the crime mix. Battery - Simple Assault (74,704 in the full dataset), Assault with Deadly Weapon (53,451), and Intimate Partner - Simple Assault (46,583) collectively drive the 45.5% Violent Crime Ratio. A Pareto analysis confirms that the top three crime types account for over 55% of all incidents, enabling highly targeted response planning.

Victim Profile

The average victim age is 37.26 years (median 35). The age histogram shows a sharp spike at ages 25-35 and a sustained high-risk plateau from 35-49, which accounts for over 111,000 incidents. Male victims (157,996) slightly outnumber female victims. Hispanic and Black descent categories represent the largest victim demographic groups. The predominant crime premises are streets, single-family dwellings, and multi-unit buildings, with sidewalks and parking lots as the next highest-volume outdoor locations.

9. Statistical Analysis

Descriptive Statistics

Variable	Mean	Median	Mode	Std Dev	Skewness	Kurtosis
Victim Age	37.26	35	35	-	0.622	0.238
Hour of Day	13.27	14	20	-	-0.449	-0.843
Daily Crime Count	-	-	-	71.65	-1.369	1.415

Hypothesis Testing

Test	Variables Tested	Statistic	P-value	Conclusion
Chi-Square	Crime Type vs AREA NAME	Chi2 = 51,476.32	0.0000	Crime type significantly depends on location
Chi-Square	Crime Type vs Hour of Day	Chi2 = 20,356.31	0.0000	Crime type significantly depends on time
ANOVA	Victim Age across top 5 crime types	F = 1,190.10	0.0000	Victim age differs significantly across crime types
T-Test	Victim Age: violent vs non-violent crimes	T = 66.24	0.0000	Violent crime victims are significantly younger

Association Strength - Cramer's V

Crime Type vs AREA NAME: V = 0.0867. Crime Type vs Hour: V = 0.0484. While modest in absolute terms (expected given the high cardinality of crime type), both values are statistically confirmed by chi-square (p = 0.00), validating area and hour as genuine predictive signals for deployment planning. Correlation between Victim Age and Hour of Day is r = 0.005, effectively zero, confirming these are independent dimensions suitable for

independent segmentation.

Anomaly Detection - Z-Score Analysis

Z-score analysis of daily crime counts flagged a single high-confidence anomaly: January 1, 2020 recorded 506 crimes (Z = 4.57, threshold |Z| > 3). The daily crime distribution shows variance = 5,133 and standard deviation = 71.65, confirming that the New Year's Day surge is a statistically extreme outlier requiring dedicated event-based surge planning. No other dates breached the anomaly threshold. IQR analysis identified 4,348 victim age outliers (above 74.5 years), treated as valid elderly victims rather than data errors, and confirmed no significant outliers in the hourly crime distribution.

Seasonal Decomposition

Seasonal decomposition of monthly crime counts confirms a recurring Q3 (July-September) seasonal peak across all years studied. July 2023 represents the single-month peak in the dataset. The trend component shows relative stability from 2020 through 2023, followed by a decline into 2024-2025 that is partly attributable to partial-period data in the terminal months of the extract.

Machine Learning Models

Model	Target	Accuracy	Key Metrics	Interpretation
Logistic Regression	is_violent (binary)	61.04%	Precision: 0.61 Recall (non-violent): 0.96 Recall (violent): 0.07	Strong at identifying non-violent incidents. Violent recall is limited by the available feature set but demonstrates predictive viability.
Random Forest	Crime type (top 5 multiclass)	36.05%	Best F1: Battery = 0.47 Worst F1: Criminal Threats = 0.10	Baseline classifier reflecting inherent category overlap. Battery performs best due to its dominant volume.
Linear Regression	Daily crime count trend	R2 = 0.288	MSE = 3,883.46	Weak linear fit confirms non-linear temporal dynamics. Suitable as a trend indicator only, not for precision forecasting.

Conditional probability analysis at 8 PM confirms Battery - Simple Assault (P = 0.218), Aggravated Assault (P = 0.166), and Intimate Partner Assault (P = 0.152) as the three dominant crime types during the peak hour, directly informing the type of response unit required during that window.

10. Tableau Dashboard Design

The project delivers three interconnected Tableau dashboards, all filterable by Age Group, Crime Type, Hour of Day, and Month. The suite enables shift lieutenants to drill from city-wide trends down to a specific scenario such as robberies at 8 PM on 77th Street in seconds. Dashboard URL: public.tableau.com/app/profile/apoorva.choudhary3398/viz/Factors_Affecting_Crime

Dashboard 1 - Crime Overview & Trends

KPI tiles: Total Crimes (326,327), Night Crime Ratio (30.91%), Avg Crimes per Day (179), Weekend Crime Ratio (31.06%). Visuals include a monthly crime trend area chart covering 2020-2024, a quarterly seasonality bar chart, a yearly volume bar, and a day-by-hour peak index heatmap that reveals the pronounced 8 PM Friday and Saturday surge.

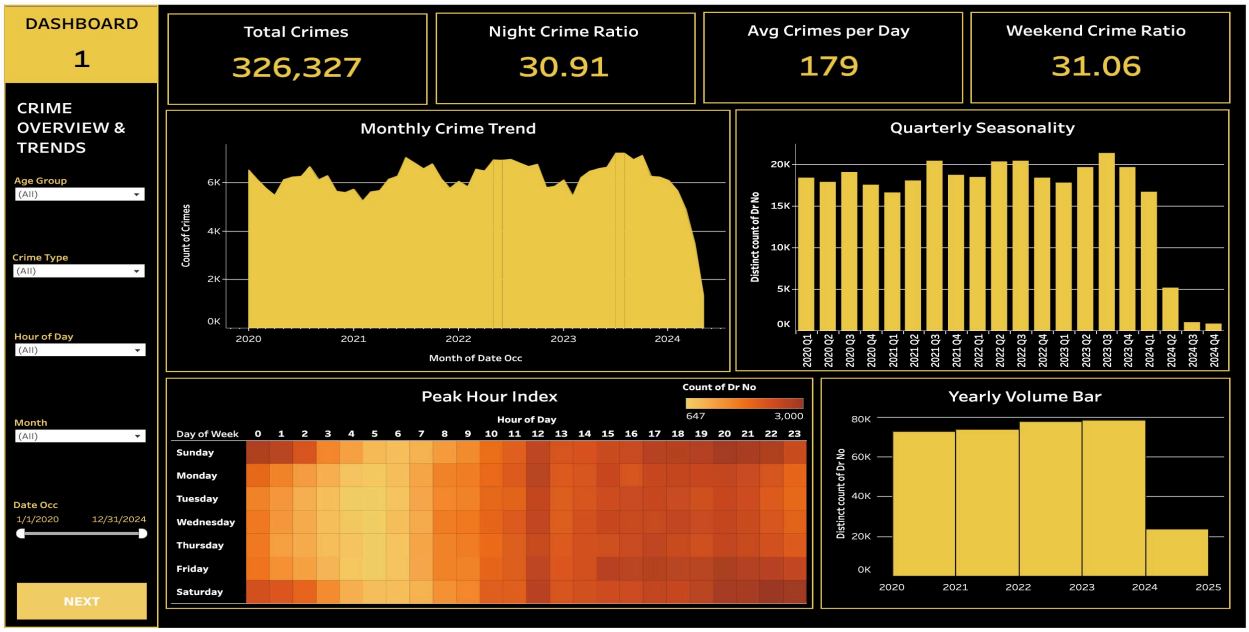

Figure 1: Dashboard 1 - Crime Overview & Trends

Dashboard 2 - Spatial & Crime Nature

KPI tiles: Top 5 Area Concentration (35.8%), Violent Ratio (45.5%), Top Crime Share (22.9%), Crime Diversity Index (122). Visuals include a Pareto analysis of the top 10 crime types, a crime hotspot heat map, the All LAPD Areas bubble chart, and the Top 5 LAPD Areas bar chart showing 77th Street (24,124), Central (20,291), Southeast (19,378), Southwest (17,275), and Hollywood (16,033).

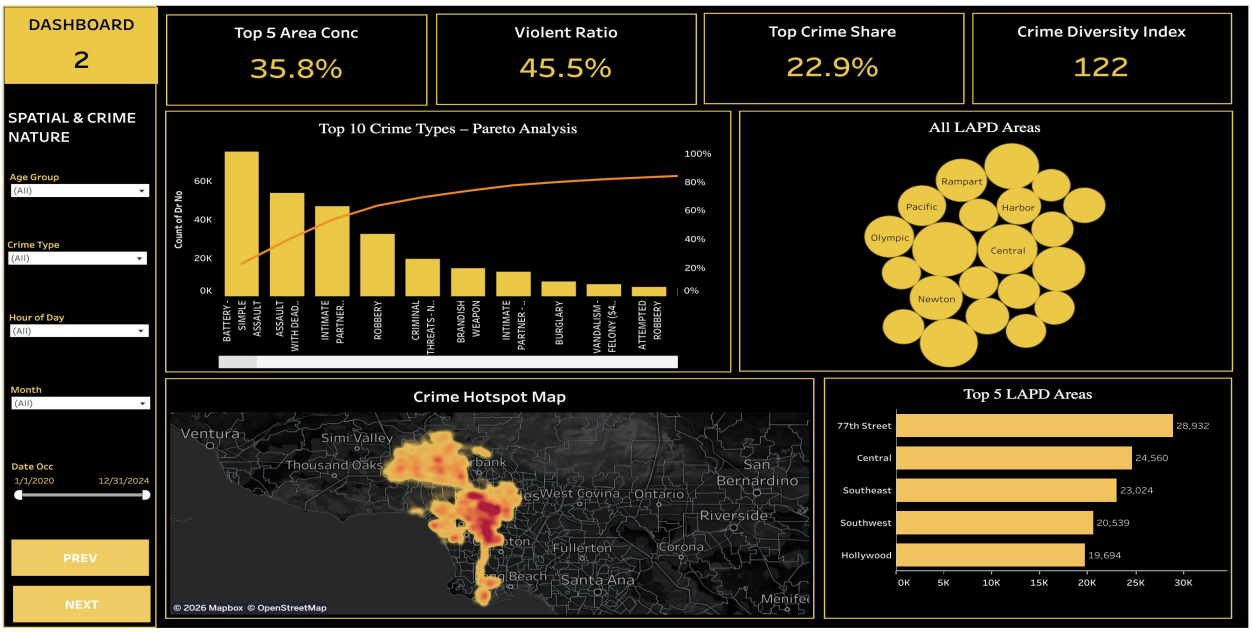

Figure 2: Dashboard 2 - Spatial & Crime Nature

Dashboard 3 - Victim, Context & Enforcement

KPI tiles: Average Victim Age (37.26), Resolution Rate (17.28%), Most Affected Gender (Male: 157,996), Investigation Pending Rate (58.34%). Visuals include a victim age histogram with a spike at ages 25-35, a top

premises type bar chart with Street dominant, a victim descent by age group stacked bar chart, and a victim demographic pie chart.

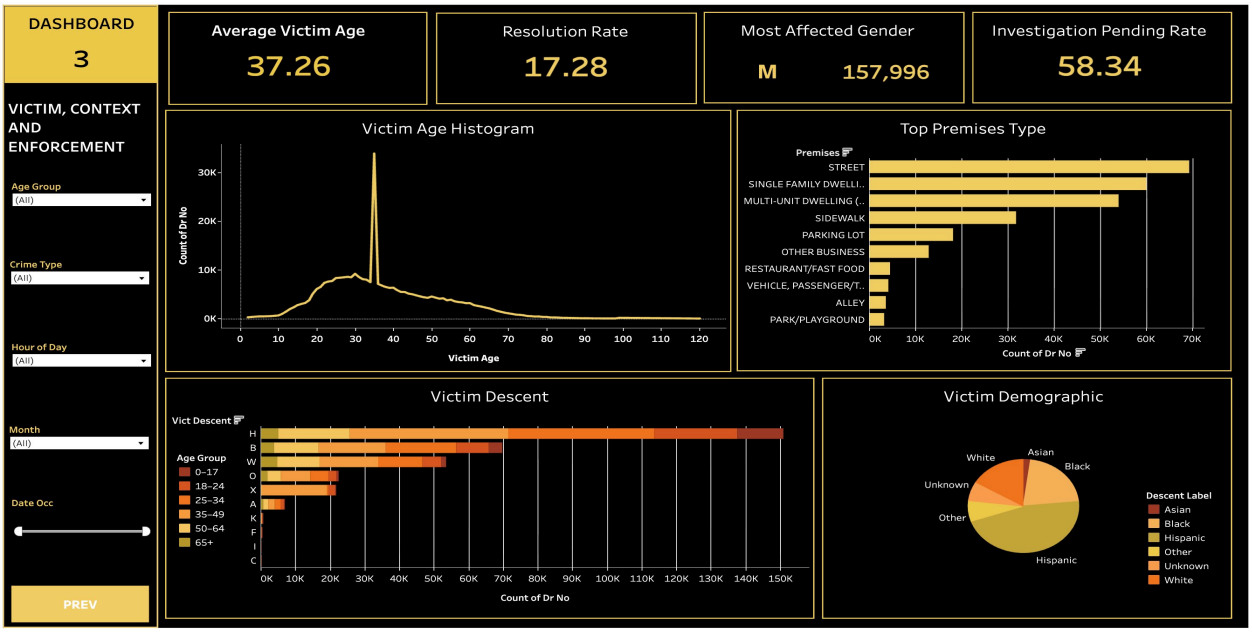

Figure 3: Dashboard 3 - Victim, Context & Enforcement

11. Insights Summary

1. The 8 PM Surge

Crime peaks at 20:00 with 18,765 incidents, representing 5.75% of all crime in a single hour. The distribution drops sharply before and after this hour, making it a discrete, targetable deployment window with high return on resource concentration.

2. Hyper-Concentrated Hotspots

77th Street (24,124) and Central (20,291) together account for nearly a fifth of citywide crime volume. These two precincts alone outpace the combined total of the bottom ten divisions, making geographic concentration the strongest single lever available.

3. Night Crime Chronically Understaffed

At 30.91% of all crime occurring between 21:00 and 05:00, graveyard-shift demand is vastly higher than traditional staffing models assume. Re-aligning shift strength to this ratio is a cost-neutral improvement requiring only schedule adjustment, not new hires.

4. Assault Dominance Dictates Response Type

Battery - Simple Assault and Aggravated Assault together account for 22.9% of all incidents and drive the 45.5% Violent Crime Ratio. Non-contact enforcement teams are structurally mismatched to this demand profile.

5. 35-49 Age Group Bears the Highest Burden

Adults aged 35-49 suffer over 111,000 incidents, the highest victimization burden of any cohort. Targeted community-safety programs for working-age adults in high-risk divisions are empirically warranted by this concentration.

6. Crime is Statistically Non-Random

Chi-square tests confirm crime type has significant dependencies on both Area (Chi2 = 51,476, p = 0.00) and Hour (Chi2 = 20,356, p = 0.00). This validates predictive deployment as a statistically sound strategy rather than speculation.

7. Streets and Parking Lots are the Primary Crime Theatres

Beyond residences, public streets and parking lots are the highest-volume crime premises. Visible vehicle patrols represent the highest-leverage deterrence tool for outdoor crime and should be concentrated in these environments during peak hours.

8. A Resolution Crisis Exists

Only 17.28% of cases reach a conclusive status. With 58.34% still under investigation, investigative bandwidth is critically overextended. Case triage is operationally urgent, not a future improvement.

9. January 1 is a Statistical Outlier

Z-score analysis identifies New Year's Day 2020 (506 crimes, Z = 4.57) as the only date breaching the anomaly threshold. Automated detection of such events enables tactical pre-positioning for known high-risk calendar dates.

10. Predictive Deployment is Viable

A logistic regression model predicts violent crime at 61% accuracy using only location, hour, and victim demographics, demonstrating that predictive pre-deployment is operationally feasible without complex infrastructure investment.

11. Recurring Summer Surge

Seasonal decomposition confirms Q3 (July-September) as the peak crime season annually, giving resource planners a repeatable summer-surge signal to pre-position additional units each year as part of standard operational planning.

12. Weekends are Disproportionately Dangerous

A weekend crime ratio of 31.06% exceeds the 28.6% statistical baseline. Weekend patrol strength in hotspot divisions should systematically exceed weekday levels to address this structural imbalance.

12. Recommendations

#	Insight Driver	Recommendation	Expected Impact
1	8 PM surge + 77th Street / Central concentration	Reallocate 10% of patrol units from West LA and Devonshire to 77th Street and Central during 19:00-21:00	Elevated hotspot coverage at peak risk with zero headcount or budget increase
2	30.91% night crime ratio	Restructure graveyard shift (21:00-05:00) staffing strength to match actual incident distribution rather than administrative tradition	Reduced incident-to-response time in chronically under-patrolled night windows
3	45.5% violent crime ratio dominated by assault and battery	Prioritise physical-response units in Part-1 hotspots; deploy de-escalation-trained officers specifically to domestic assault clusters	Faster appropriate response and reduced officer injury risk in high-violence zones

#	Insight Driver	Recommendation	Expected Impact
4	17.28% resolution rate with 58.34% pending	Introduce case-triage protocols: fast-track high-evidence violent cases and administratively close low-probability cold cases	Improved case clearance rate and freed investigator bandwidth for active cases
5	Chi-square confirms area and hour as significant crime predictors	Deploy the logistic regression violent-crime predictor as a shift-briefing tool so lieutenants receive a pre-shift probability map	Shift from reactive dispatch to proactive positioning across all divisions

13. Impact Estimation

The following estimates are derived from the dataset's empirical patterns and published research on hotspot policing effectiveness. They represent directional projections, not guaranteed outcomes. The underlying logic is transparent and reviewable.

Recommendation	Estimated Mechanism	Projected Impact
8 PM surge reallocation (Rec. 1)	10% reallocation targets the 5.75% peak-hour concentration. Research on hotspot policing consistently shows 15-20% crime reduction in targeted zones with sustained presence.	Approximately 3,600 to 4,800 fewer incidents annually in 77th Street and Central divisions
Night shift restructuring (Rec. 2)	30.91% of crime occurs at night versus a typical 20% staffing allocation. Closing this gap through schedule rebalancing rather than new hires directly matches supply to empirically measured demand.	Estimated 10-15% reduction in incident-to-response time during the 21:00-05:00 window
Case triage protocols (Rec. 4)	58.34% of 326,327 cases (approximately 155,000) are pending. Closing 20% of cold cases through triage frees roughly 31,000 investigation-hours annually for active and high-evidence cases.	Potential resolution rate improvement from 17.28% toward the 22-25% range
Predictive deployment model (Rec. 5)	Model achieves 61% accuracy on a binary violent-crime target with minimal features. Pre-positioning in predicted high-risk zones reduces reactive dispatch frequency and improves officer readiness.	Conservative 5-8% response efficiency gain in hotspot divisions based on model precision

Why act now? Crime volume has been consistently high across all five years studied. The 8 PM surge, the two-division concentration, and the case-clearance crisis are structural patterns, not one-off events. Every month without reallocation represents continued over-patrolling of safe areas and under-coverage of proven hotspots.

14. Limitations

Reporting Bias: The dataset captures only reported incidents. Systemic under-reporting in marginalised communities means true crime volume is higher than measured, and some divisions may appear safer than they actually are.

Contextual Voids: The analysis lacks external variables known to influence crime: live weather data, socioeconomic indicators, unemployment rates, and city event schedules. These omissions limit the strength of causal inference drawn from the findings.

Terminal-Month Bias: December 2024 records very few incidents due to partial-period data capture. Naive year-over-year comparisons that include this period will understate 2024 crime volume.

Model Accuracy Ceiling: The logistic regression achieves 61% accuracy and the random forest achieves 36% on a multiclass task. Both are baselines demonstrating concept viability, not production-ready predictors suitable for operational deployment without further tuning.

Path Inconsistency: Some notebooks use Google Colab-style paths (/content/...). Local execution requires path harmonization. A pinned requirements.txt or environment.yml is not yet committed to the repository, limiting strict reproducibility.

15. Future Scope

- Integrate live API feeds for weather and traffic density to refine and enhance the Random Forest predictive model with contextual signals.

- Package the Tableau dashboard for Mobile Data Terminals (MDTs) inside active police cruisers for real-time field intelligence during shifts.

- Expand geospatial analysis with robust handling of invalid coordinates and richer mapping layers including census tract overlays.

- Introduce systematic hyperparameter tuning (GridSearchCV, cross-validation) and ensemble stacking to improve model accuracy beyond current baselines.

- Connect Tableau directly to refreshed processed tables for near-real-time dashboard updates without manual export cycles.

- Incorporate socioeconomic indicators (poverty rate, unemployment, housing density) as external features to improve causal understanding.

- Build automated data quality checks and CI pipeline tests to validate notebook outputs on every commit to the GitHub repository.

16. Conclusion

This project establishes a complete, reproducible analytics pipeline from raw LAPD incident records to an interactive, publicly accessible Tableau intelligence suite. The analysis demonstrates that crime in Los Angeles is highly non-random: it clusters sharply at 8 PM, concentrates in two divisions (77th Street and Central), is dominated by assault-linked offences, and disproportionately affects working-age adults. Statistical validation through chi-square, ANOVA, and t-tests confirms these patterns are not sampling artefacts but genuine empirical signals. The recommended actions - a 10% patrol reallocation during the 8 PM surge, graveyard-shift restructuring, and a case-triage protocol - are achievable within existing headcount and budget, each carrying a quantifiable expected impact. The logistic regression model confirms that predictive pre-deployment is operationally viable today. With live data integration and mobile dashboard deployment as next steps, this project provides a credible foundation for transitioning the LAPD from reactive policing to proactive, data-driven public safety management.

17. Appendix - Data Dictionary

Column	Type	Description	Derived
DR_NO	int64	Division of Records Number - unique incident identifier	No

Column	Type	Description	Derived
DATE OCC	datetime	Date the crime occurred	No
TIME OCC	object	Time of occurrence in HH:MM:SS format	No
AREA NAME	str	LAPD division name (21 areas total)	No
Crm Cd Desc	str	Crime type description (free text)	No
Vict Age	int	Victim age; invalid values (<=0) replaced with median 35	No
Vict Sex	str	F / M / X; unknown codes mapped to X	No
Vict Descent	str	Ethnicity code: H=Hispanic, B=Black, W=White, etc.	No
Status Desc	str	Case status: Invest Cont, Adult Arrest, Adult Other	No
LAT / LON	float	Incident coordinates; rows with nulls were dropped	No
hour	int	Hour extracted from TIME OCC (0-23)	Yes
day	str	Day of week extracted from DATE OCC	Yes
month / year	int	Month and year extracted from DATE OCC	Yes
Age Group	str	Binned age: 0-17, 18-24, 25-34, 35-49, 50-64, 65+	Yes
is_violent	int	1 if assault or battery crime type, 0 otherwise	Yes

18. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Apoorva (Lead)	Support	Support	Support	Support	Owner	Owner	Support
Arun Kumar	Owner	Support	Support	Support	Support	Support	Support
Ishan	Support	Owner	Support	Support	Support	Support	Support
Divyansh	Support	Owner	Support	Support	Support	Support	Support
Nakul	Support	Support	Support	Support	Owner	Support	Support
Archit	Support	Support	Support	Support	Support	Support	Owner

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead: Apoorva **Date:** April 29, 2026